

Debunking Fake News in Online Social Networks without Text Analysis

Xing Su, Jian Yang, Jia Wu and Zitai Qiu

School of Computing, Macquarie University, Sydney, Australia

Email: {xing.su2, zitai.qiu}@students.mq.edu.au, {jian.yang, jia.wu}@mq.edu.au

Abstract—Since the inception of online fake news detection, the technique of natural language processing has predominantly been leading the field by utilizing text classification to discern veracity. From the network perspective, news traveling within a social network typically exhibits non-textual correlations aligned with the network of news propagation or news-user interaction. Therefore, with the advancement of graph learning, there have been emerging approaches incorporating graphs of social contexts as auxiliary information, of which the performance still relies on learning semantics from news text. As fake news becomes more adept at employing the writing pattern of real news and the assessment of certain news contents requires domain-specific knowledge, distinguishing real news from fake ones based on the text has become increasingly challenging. This raises a question: Can we debunk fake news without going through the text? Thus, this work aims to explore the feasibility of differentiating between real and fake news by capturing its relationships with other news and people in the network. We propose a method named ComE-DeFake which extracts intricate relations beyond pairwise of news and users in social contexts to detect fake news. Experimental results reveal that our method without using news text outperforms all baseline methods. This suggests that, if high-order complicated relations are fully captured, it is achievable to debunk fake news without analyzing its text.

Index Terms—Fake News Detection, Hypergraph, High-order Relations, Text-independent

I. INTRODUCTION

Nowadays, social media such as Facebook and Twitter have emerged as the primary platforms for people to obtain news and share information. While these platforms offer instant access to real-time news from diverse perspectives, the ease of engagement and interaction has led to soaring of fake news. Therefore, it is an imminent task to debunk online fake news to ensure the public receives trustworthy information and guidance with utmost reliability. Various methods have been developed for debunking fake news in online social networks [1], [2]. A popular technique is to use text classification [3], but these approaches have limited performance as they only consider news text and overlook the social context [4]–[6]. With the development of graph learning, some models incorporate social relations to detect fake news [7], but these methods still rely heavily on learning the semantics of news text while graph plays an auxiliary role.

Online fake news is becoming more adept at disguising itself by emulating writing patterns of real news. For example, in

the PolitiFact dataset shown in Fig. 1, many high-frequency words appear in both real and fake news, including terms like “obama”, “trump”, and “president”. Such imitation leads to fake news and real news becoming very similar, and their textual representations become less discriminative. This is especially problematic in the case of short news, such as tweets, where the text provides limited semantic information for discerning authenticity (**Challenge 1**). Furthermore, assessing many news requires domain-specific knowledge beyond the general public’s grasp (**Challenge 2**) [8]. For instance, discerning a piece of fake news about COVID-19, such as “If you have COVID and aren’t showing symptoms you won’t transmit the disease,” requires medical expertise.

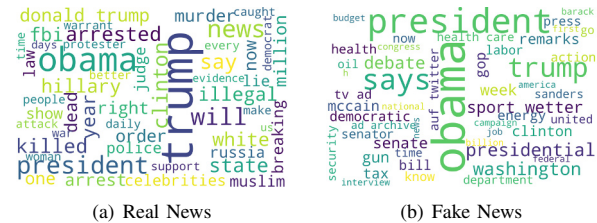


Fig. 1. High-frequency Words of Real News and Fake News in PolitiFact Dataset.

These aforementioned circumstances contribute to the growing challenge of conducting fake news detection through news text analysis. Thus, Rosenfeld *et al.* [9] revealed a crucial insight: misinformation can be accurately discerned by solely analyzing propagation patterns, without considering any textual information. Based on this key finding, our work aims to propose a model that detects fake news without news text analysis. Given the good auxiliary performance of graph learning in existing methods and the inherent network characters of news that involve propagation and interaction with users, this work constructs a social graph to reflect intricate relations between news and users. The objective is to explore the possibility of discerning real and fake news solely through structural relations among news and users.

To achieve this goal, employing the existing methods which construct simple graphs does not yield satisfactory results, since they only depict pairwise relations and capture limited information. Suppose a network of users participating in news propagation, as shown in Fig. 2 (a). We can observe that users are generally active in community form and propagate news in groups, such as u_1 , u_2 , u_3 and u_4 collectively

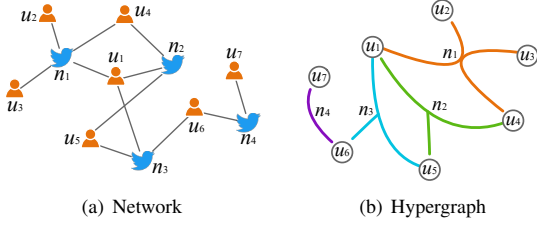


Fig. 2. Hypergraph Construction by Relation of News and Users. (a) A network of users participating in the news. (b) Hypergraph: nodes represent users while each different-colored hyperedge represents a piece of news.

spreading news n_1 , but existing pairwise-oriented methods find it challenging to efficiently capture community-level patterns for fake news detection. Due to the positive correlation between news authenticity and user credibility [10], [11], the credibility of a user community participating in a piece of news directly impacts the authenticity of that news. Capturing such community-level relations brings the opportunity to debunk fake news without considering news text.

Thus, we employ the concept of hypergraph to capture such community-level relations between news and users. Hypergraphs are generalizations of simple graphs that can connect an arbitrary number of nodes in an edge and extract high-order relations among them [12]. As shown in Fig. 2 (b), we construct a hypergraph by extracting users as nodes and news as hyperedges. Each hyperedge can connect more than two nodes representing a group of users spreading a piece of news. For instance, the nodes of u_1 , u_2 , u_3 , and u_4 are extracted on the hyperedge of n_1 . Such construction captures not only the underlying relation among diverse news that is connected due to the same users' participation, but also the community-level interaction between a piece of news and the users who participated in its dissemination.

In this work, we propose a method based on **Community-driven hypergraph learning** to learn high-order relations for **Debunking Fake** news in online social networks, abbreviated as ComE-DeFake. Specifically, we first utilize hyperedge convolution to capture patterns of news dissemination, so that the high-order correlations can be perceived. Then a community-driven encoding mechanism is proposed to depict complicated interaction relations among users, where each user representation embeds credibility with credible or suspicious community membership. Subsequently, a high-order relation augmentation component is designed to aggregate the credibility of involved users for each piece of news. Thus, we acquire the discriminative news representations augmented by high-order relation, followed by a fake news detector to identify the real and fake news. The main contributions of this work are as follows:

- To our best knowledge, this is the first attempt to uncover fake news without relying on its text. This work addresses the challenge of requiring domain-specific knowledge to discern certain news and also mitigates the issue of insufficient semantic information in short news in social

networks. Experimental results validate the feasibility of debunking fake news without its text analysis, even at the early stage of news publication and propagation.

- By constructing a social hypergraph of news and users, we tackle the limitations of existing methods which only capture pairwise relations. Experiments demonstrate that our method successfully depicts beyond pairwise relations to enhance fake news detection.
- ComE-DeFake inventively integrates community detection to debunk fake news, providing community-level user credibility for augmenting news representation through user clustering. Experiments illustrate that such innovative incorporation improves the effectiveness of our method even without exploiting news text.

II. RELATED WORK

A. Hypergraph Learning

Since the traditional graph learning methods primarily focus on pairwise connections [13], hypergraph learning has recently emerged to model high-order relations beyond pairwise associations. The first work that applies GNNs to hypergraphs is HGNN [14], which designs a hyperedge convolution operation. HyperGCN [15] trains a GCN on hypergraph by spectral theory. Hyper-SAGNN [16] develops a self-attention based GNN for hypergraphs. HyperGCL [17] is a hypergraph generative model. HcoN [18] reconstructs the hypergraph collaboratively. Although their capacity to capture complicated high-order relations is suitable for complex social networks, their application in debunking fake news in online social networks is relatively limited. HyperGAT [19] is a text classification method that captures high-order relations in texts by constructing hypergraphs of textual contents and ignores social contexts. HGFND [20] and HG-SL [21] are the hypergraph-relevant methods for fake news detection, but they are both semantics-dependent.

B. Fake News Detection

From the perspective of news contents, text-based approaches detect fake news by distinguishing linguistic features and writing styles between real and fake news [3], [22]. Apart from analyzing news text itself, auxiliary textual information is also integrated to enhance fake news detection, such as comments [4], [5], emotions [23], and stances [24]. While text-based models utilize textual information and auxiliary features for fake news detection [25], they commonly analyze news piece by piece, overlooking the intrinsic news-spreading mechanism. As a result, they may neglect valuable news propagation information. These methods also lack user information which is crucial for understanding social context to accurately identify fake news. Consequently, the performance of text-based approaches is limited in the social network context.

Due to the propagation nature of news aligning precisely with the structural property of graphs, graph learning shows good results in fake news detection [26], [27]. Researchers typically extract structural relations of words or sentences [28], construct news propagation graphs [6], [7] or incorporate

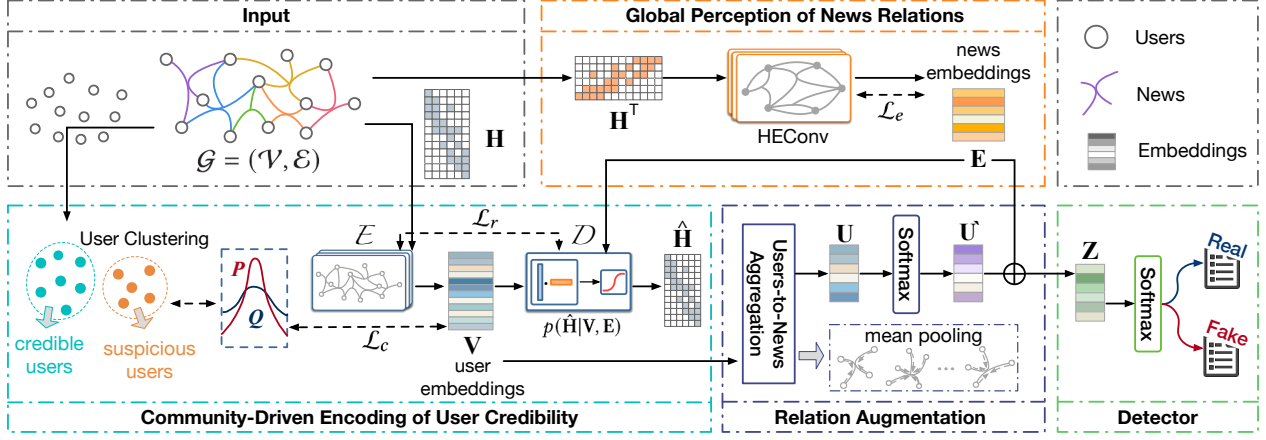


Fig. 3. The Framework of ComE-DeFake (E : high-order credibility encoder, D : collaborative decoder).

social contexts [27], [29], [30] to detect fake news from both textual and structural perspectives. To fully harness the potential of graph structure for accurate results, researchers also incorporate structured information, such as knowledge graphs [31], [32], heterogeneous information networks [33]–[35], and user information [10], [36], [37]. Although the variety of methods considers user information in social contexts, they are inadequate in modeling the high-order relations between news and users. Furthermore, the aforementioned methods to varying degrees rely on the news text and cannot solely determine the news authenticity through its network structure. We aim to address these problems in this paper.

III. PROBLEM STATEMENT

1) *Notations and Definition*: Let $\mathcal{N} = \{n_1, n_2, \dots, n_t\}$ denote the news set, where t represents the number of news. $\mathbf{X}_e$ represents the textual attributes of news. Let $\mathcal{U} = \{u_1, u_2, \dots, u_m\}$, where m is the number of users. The corresponding user attributes are defined as $\mathbf{X}_u$.

A hypergraph is a generalized graph, it can be formulated as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ indicates the node set. $\mathcal{E}$ signifies the set of hyperedges, where each hyperedge connects to a subset of nodes rather than two nodes. $\mathbf{H} = |\mathcal{V}| \times |\mathcal{E}|$ denotes incidence matrix, where $\mathbf{h}(v, e) = 1$ if the node v is on hyperedge e (i.e., $v \in e$), else $\mathbf{h}(v, e) = 0$.

2) *Hypergraph Construction*: To model the global relation between diverse news and the users involved in their propagation in online social networks for fake news detection, this work defines an attributed hypergraph as $\mathcal{G} = (\mathcal{U}, \mathcal{E}, \mathbf{X}_u, \mathbf{X}_e)$ to represent the high-order relations. Here the node set $\mathcal{U}$ denotes the users who are involved in news propagation. The hyperedge set $\mathcal{E}$ denotes news, where each hyperedge e_i represents a piece of news n_i and can connect more than two users who are involved in the spread of this news. $\mathbf{X}_u$ and $\mathbf{X}_e$ represents attributes of nodes (i.e., users) and hyperedges (i.e., news), respectively.

IV. THE PROPOSED MODEL

This section introduces our proposed method in detail. Fig. 3 illustrates the overall framework of ComE-DeFake, which consists of four main components: 1) Global perception of news relations, which learns high-order and global representations of news through hyperedge learning; 2) Community-driven encoding of user credibility, capturing community-level credibility representations of users through a hypergraph autoencoder with the community layer; 3) Users-to-News Relation Augmentation, which integrates user credibility into their involved news to augment news representations; and 4) Detector, a classifier for identifying real and fake news.

A. Global Perception of News Relations

Social media is in essence a vast interconnected network, within which various news exhibits intricate structural relations. If the complicated structural connections of news beyond pairwise can be globally perceived, it will provide more macro high-order information for accurate fake news detection. To perceive the intricate high-order relations in news, we focus on hyperedges representing news in our constructed hypergraph and employ a hypergraph convolution network [14] for hyperedge learning of the news. Since the standard convolution is defined on nodes as formulated in Eq. (3), the nodes and hyperedges in $\mathbf{H}$ are treated as hyperedges and nodes in $\mathbf{H}^\top$ to filter hyperedges. We formulate the convolution operation on the hyperedges as follows:

$$\mathbf{E}^{(l)} = \sigma(\mathbf{B}_e^{-\frac{1}{2}} \mathbf{H}^\top \mathbf{J} \mathbf{B}_v^{-1} \mathbf{H} \mathbf{B}_e^{-\frac{1}{2}} \mathbf{E}^{(l-1)} \Theta_e^{(l-1)}), \quad (1)$$

where $\mathbf{J}$ signifies the learnable parameter during the training process, $\mathbf{B}_e$ and $\mathbf{B}_v$ are the diagonal degree matrix of hyperedges and nodes, respectively. Θ_e is the filter, and σ indicates the nonlinear activation function. $\mathbf{E}^{(0)} = \mathbf{X}_e$. In ComE-DeFake, $\mathbf{X}_e$ is the identity matrix $\mathbf{I}$, when we debunk fake news without considering its text.

For each piece of news, *i.e.*, each hyperedge, the objective of this component is to minimize the cross-entropy loss:

$$\mathcal{L}_e = - \sum_{y_i \in \mathcal{Y}_{tr}} y_i \log \hat{y}_i - (1 - y_i) \log(1 - \hat{y}_i), \quad (2)$$

where $y_i \in \mathcal{Y}_{tr}$ indicates that the ground truth label of news n_i is from the training data label set $\mathcal{Y}_{tr}$. $y_i = 1$ represents that the news label is fake, otherwise it is real. $\hat{y}$ denotes the predicted value.

B. Community-Driven Encoding of User Credibility

Since news creation and dissemination in social networks originate from users, we treat users as nodes when constructing the hypergraph. This construction enables us to abstract high-order relations between users who are connected by the news with which they collectively engage. To this end, we employ a hypergraph autoencoder to learn the high-order interaction between users to reflect their credibility, so as to employ the credibility to make news representations more discriminative. Considering that users in online social networks commonly have community properties [38], and the positive correlations between user credibility and news authenticity [10], [11], we divide users engaged with news into credible users likely to propagate real news and suspicious users who may maliciously share fake news. By introducing a community layer into the encoder of the hypergraph autoencoder, the learned user credibility can be optimized.

1) *High-Order Credibility Encoder*: To encode high-order interactions between users, we use the aforementioned hypergraph convolution operating on nodes. It performs node-edge-node transform, which depicts relations between users connected by the same news pieces and refines the credibility-related user attributes by capturing user-news-user relations. The convolution is defined as:

$$\mathbf{V}^{(l)} = \sigma(\mathbf{D}_u^{-\frac{1}{2}} \mathbf{H} \mathbf{W} \mathbf{D}_e^{-\frac{1}{2}} \mathbf{H}^\top \mathbf{D}_u^{-\frac{1}{2}} \mathbf{V}^{(l-1)}) \Theta_u^{(l-1)}, \quad (3)$$

where $\mathbf{W}$ signifies the learnable parameter during the training process. $\mathbf{V}^{(0)} = \mathbf{X}_u$. $\mathbf{D}_u$ and $\mathbf{D}_e$ are the diagonal degree matrix of nodes and hyperedges, respectively.

2) *Collaborative Decoder*: Because the purpose of this work is to investigate the effectiveness of high-order structural relations in distinguishing between real and fake news without considering news text (*i.e.*, hyperedge attributes), reconstructing node or hyperedge attributes may not align well with our aim [18]. Through encoding hyperedges and nodes in the previous procedures, we capture the high-order latent representations of news and users, respectively. To leverage both representations collaboratively, we reconstruct the incidence matrix $\mathbf{H}$, which serves to establish the connections between news and users. Specifically, the decoder recovers the relations across nodes and hyperedges through the latent representations of users and news as follows:

$$p(\hat{\mathbf{H}}|\mathbf{V}, \mathbf{E}) = \prod_{v \in V} \prod_{e \in E} p(\hat{\mathbf{H}}(v, e) | \mathbf{V}(v), \mathbf{E}(e)), \quad (4)$$

where $\mathbf{V}, \mathbf{E}$ are the learned representations of nodes and hyperedges, respectively. $p(\hat{\mathbf{H}}(v, e) | \mathbf{V}(v), \mathbf{E}(e)) =$

$\text{sigmoid}(\mathbf{V}(v) \mathbf{E}(e)^\top)$. Hence, the hypergraph reconstruction loss function is minimized by:

$$\mathcal{L}_r = \mathbb{E}_{\mathbf{V}, \mathbf{E}} [\log p(\hat{\mathbf{H}} | \mathbf{V}, \mathbf{E})]. \quad (5)$$

3) *User Community Self-Optimization*: To make the representations of credible and suspicious users more distinguishable in a lower-dimensional space and leverage informative user credibility for reinforcing fake news detection, we introduce a community layer when encoding user high-order relations. This layer optimizes user representations at a community level in the reconstruction training process. The hidden representations of nodes are self-optimized by minimizing a clustering objective as follows:

$$\mathcal{L}_c = \text{KL}(P || Q) = \sum_i \sum_c p_{ic} \log \frac{p_{ic}}{q_{ic}}, \quad (6)$$

where $\text{KL}(\cdot || \cdot)$ is Kullback-Leibler divergence between two distributions, Q is the distribution of soft community labels, P is the target distribution of Q . q_{ic} measures the similarity between node embedding v_i and cluster center embedding μ_c as follows:

$$q_{ic} = \frac{(1 + \|v_i - \mu_c\|^2)^{-1}}{\sum_k (1 + \|v_i - \mu_k\|^2)^{-1}}, \quad (7)$$

where k is the number of communities. p_{ic} is the target distribution defined as: $p_{ic} = \frac{q_{ic} / \sum_i q_{ic}}{\sum_k (q_{ik} / \sum_i q_{ik})}$. The community loss forces Q to approach P , so as to set these credible or suspicious communities as soft labels to supervise Q 's representation learning.

C. Users-to-News Relation Augmentation

Each hyperedge can connect multiple nodes, indicating that a piece of news is being propagated by a group of users. To facilitate the assistance of user credibility in fake news detection, we design a user-news relation augmentation to amplify the news representations by performing nodes-to-hyperedge aggregation. Specifically, we define $\mathcal{U}_i^s \subseteq \mathcal{U}$ to represent a subset of users involved in the news n_i . To obtain the representative embedding of these users $\mathcal{U}_i^s$, we aggregate credibility embeddings of users involved in news n_i by mean pooling. Then, the news representation of e_i is added with the representative user embedding after a softmax layer. The process can be expressed as follows:

$$\mathbf{u}_i = \text{MEAN-POOLING}\{\mathbf{v}_j, \forall j \in \mathcal{U}_i^s \subseteq \mathcal{U}\}, \quad (8)$$

$$\mathbf{u}'_i = \text{SOFTMAX}(\mathbf{u}_i), \quad (9)$$

$$\mathbf{z}_i = \text{ADDITION}(\mathbf{e}_i, \mathbf{u}'_i) = \mathbf{e}_i \oplus \mathbf{u}'_i, \quad (10)$$

where $\mathbf{v}_j$ is the credibility embedding of j -th user in $\mathcal{U}_i^s$, and $\mathbf{u}_i$ is the aggregated credibility embedding of users involved in the dissemination of news n_i . $\mathbf{u}'_i$ is the normalized $\mathbf{u}_i$. After adding operation with hyperedge embedding $\mathbf{e}_i$, we obtain the augmented news representation $\mathbf{z}_i$.

D. Detector

After obtaining the final news representations that fuse news high-order structural information with community-level user credibility information, such integrated representations can be regarded as augmented hyperedge embeddings, because each hyperedge represents an individual news. The task of debunking fake news becomes a hyperedge classification. In ComE-DeFake, the integrated hyperedge embeddings directly pass through a softmax layer for news prediction, which is formulated as follows:

$$\hat{\mathcal{Y}} = \text{SOFTMAX}(\mathbf{Z}), \quad (11)$$

where $\mathbf{Z}$ is the output representations of news, $\hat{\mathcal{Y}}$ represents the predicted labels. In our model, we optimize the hyperedge learning, node learning with a community layer, and define the total loss function to be minimized as:

$$\mathcal{L} = \mathcal{L}_e + \alpha \mathcal{L}_r + \beta \mathcal{L}_c, \quad (12)$$

where α and β are the coefficients controlling the balance of three objectives. The training process of ComE-DeFake is summarized in Algorithm 1.

Algorithm 1 The training process of ComE-DeFake

Input: $\mathcal{G} = (\mathcal{U}, \mathcal{E}, \mathbf{X}_u, \mathbf{X}_e)$: hypergraph of news and users; $\mathcal{Y}_{tr}$: training labels of news on hyperedges.

Parameters: α, β : coefficients balancing the loss function.

Output: $\mathbf{Z}$: the learned news representations on hyperedges.

- 1: Initialization of community layer for users by K-means.
 - 2: **while** epoch **do**
 - 3: Update $\mathbf{E}$ of hyperedges by Eq. (1).
 - 4: Update $\mathbf{V}$ of nodes by Eq. (3).
 - 5: Reconstruction of hypergraph $\hat{\mathcal{H}}$ by Eq. (4).
 - 6: Compute the cross-entropy loss $\mathcal{L}_e$ by Eq. (2).
 - 7: Compute the reconstruction loss $\mathcal{L}_r$ by Eq. (5).
 - 8: Compute the community loss $\mathcal{L}_c$ by Eq. (6).
 - 9: Backward propagation via total loss $\mathcal{L}$ by Eq. (12).
 - 10: **end while**
 - 11: **return** $\mathbf{Z}$ for classification.
-

V. TIME COMPLEXITY

Regarding the global perception of news relations, its time complexity is $\mathcal{O}(mtd_e d)$, where d_e and d are the input and hidden dimensions of hyperedges, respectively. In the community-driven encoding of user credibility, the complexity of the high-order credibility encoder is $\mathcal{O}(mtr_n r)$, the complexity of the collaborative decoder is $\mathcal{O}(mto)$, and the complexity of community layer is $\mathcal{O}(2m + m \log m)$. r_n and r are the input and hidden dimensions of nodes, while o is the output dimension. The time complexity of users-to-news relation augmentation is $\mathcal{O}(m)$. Thus, the overall time complexity of ComE-DeFake is $\mathcal{O}(m(t(d_e d + tr_n r + o) + \log m))$.

VI. EXPERIMENTS

In this section, we evaluate the performance of ComE-DeFake¹ by presenting the experimental results and investigating the following questions:

- **Q1** Does ComE-DeFake outperform all baseline methods?
- **Q2** Does ComE-DeFake benefit from each component?
- **Q3** Is ComE-DeFake robust?
- **Q4** Why ComE-DeFake performs good?
- **Q5** Is ComE-DeFake applicable in the early stages of news dissemination?
- **Q6** Is ComE-DeFake efficient?

A. Experimental Setup

1) *Datasets*: To investigate the efficacy of ComE-DeFake on fake news detection, we choose four real-world datasets, named PolitiFact [39], ReCOvery [40], MM-COVID [41], and Gossipcop [39]. These datasets cover multiple domains, with PolitiFact focusing on political news, ReCOvery and MM-COVID being related to COVID-19, while Gossipcop collects entertainment news. We used Twitter API² to crawl social contexts. TABLE I summarizes the dataset statistics, where we abbreviate ReCOvery as ReCOV and MM-COVID as MM-C.

TABLE I
THE DATASET STATISTICS.

Datasets	Nodes (Users)	Hyperedges (News)	Labels
PolitiFact	27,682	635	R: 344 / F: 291
ReCOV	36,741	2,026	R: 1,364 / F: 662
MM-C	18,672	3,992	R: 1,894 / F: 2,098
Gossipcop	72,083	10,084	R: 5,444 / F: 4,640

2) *Metrics and Baselines*: To assess the performance of our proposed ComE-DeFake and baseline methods, we use four widely-used metrics: Accuracy (ACC), Precision (Pre), Recall (Rec), and F1 score (F1). Here is a brief introduction to ten baseline algorithms: **TextCNN** [42] is a sentence classification method based on CNN (convolutional neural network). **HAN** [43] is an attention network that classifies documental content hierarchically. **BERT** [44] is a pre-trained language model for natural language processing tasks. **TextGCN** [28] extends GCN (graph convolutional network) [45] for text classification. **HyperGAT** [19] is a text data inductive method based on hypergraphs. **DualEmo** [23] distinguishes fake news by mining dual emotions from publishers and users. **HGFND** [20] constructs a hypergraph to represent group-wise interaction among news for fake news detection. **BERTweet** [46] is a large-scale language model specifically pre-trained for English Tweets. **Llama 2** [47] from Meta is a popular collection of pre-trained and fine-tuned large language models. **HG-SL** [21] globally and locally learns user spreading behavior through hypergraph learning for fake news detection.

¹Code of ComE-DeFake is available at: <https://github.com/xingsumq/ComE-DeFake>

²<https://developer.twitter.com/en/docs/twitter-api>

TABLE II
RESULTS ON THE TEST DATASETS: MEAN ACCURACY $\pm$ STANDARD DEVIATION. **COME-DeFAKE** REPRESENTS OUR METHOD WITHOUT CONSIDERING TEXT. THE BEST RESULTS ARE HIGHLIGHTED IN GRAY.

Data	Metric	TextCNN	HAN	BERT	TextGCN	HyperGAT	DualEmo	HGFND	BERTweet	Llama 2	HG-SL	ComE-DeFake
PolitiFact	ACC	0.51 $\pm$ 0.04	0.52 $\pm$ 0.02	0.79 $\pm$ 0.03	0.74 $\pm$ 0.03	0.87 $\pm$ 0.02	0.83 $\pm$ 0.03	0.88 $\pm$ 0.03	0.93 $\pm$ 0.04	0.91 $\pm$ 0.02	0.9 $\pm$ 0.01	0.98 $\pm$ 0.03
	Pre	0.31 $\pm$ 0.07	0.42 $\pm$ 0.05	0.87 $\pm$ 0.03	0.74 $\pm$ 0.03	0.87 $\pm$ 0.02	0.83 $\pm$ 0.03	0.92 $\pm$ 0.01	0.93 $\pm$ 0.04	0.92 $\pm$ 0.02	0.91 $\pm$ 0.01	0.99 $\pm$ 0.02
	Rec	0.53 $\pm$ 0.12	0.53 $\pm$ 0.06	0.81 $\pm$ 0.04	0.74 $\pm$ 0.03	0.87 $\pm$ 0.02	0.83 $\pm$ 0.03	0.89 $\pm$ 0.05	0.92 $\pm$ 0.04	0.91 $\pm$ 0.02	0.9 $\pm$ 0.02	0.98 $\pm$ 0.04
	F1	0.38 $\pm$ 0.07	0.47 $\pm$ 0.05	0.84 $\pm$ 0.02	0.74 $\pm$ 0.03	0.87 $\pm$ 0.02	0.82 $\pm$ 0.03	0.88 $\pm$ 0.03	0.93 $\pm$ 0.05	0.91 $\pm$ 0.02	0.89 $\pm$ 0.02	0.98 $\pm$ 0.03
ReCOV	ACC	0.41 $\pm$ 0.04	0.39 $\pm$ 0.03	0.76 $\pm$ 0.02	0.7 $\pm$ 0.02	0.65 $\pm$ 0.01	0.82 $\pm$ 0.03	0.92 $\pm$ 0.01	0.83 $\pm$ 0.01	0.94 $\pm$ 0.01	0.93 $\pm$ 0.01	0.98 $\pm$ 0.01
	Pre	0.27 $\pm$ 0.06	0.22 $\pm$ 0.03	0.78 $\pm$ 0.02	0.69 $\pm$ 0.08	0.58 $\pm$ 0.01	0.78 $\pm$ 0.04	0.92 $\pm$ 0.01	0.82 $\pm$ 0.02	0.93 $\pm$ 0.01	0.95 $\pm$ 0.01	0.98 $\pm$ 0.01
	Rec	0.69 $\pm$ 0.02	0.68 $\pm$ 0.07	0.75 $\pm$ 0.03	0.62 $\pm$ 0.06	0.58 $\pm$ 0.01	0.76 $\pm$ 0.04	0.91 $\pm$ 0.01	0.75 $\pm$ 0.03	0.92 $\pm$ 0.01	0.9 $\pm$ 0.02	0.97 $\pm$ 0.02
	F1	0.38 $\pm$ 0.07	0.34 $\pm$ 0.04	0.75 $\pm$ 0.01	0.59 $\pm$ 0.09	0.58 $\pm$ 0.01	0.77 $\pm$ 0.04	0.92 $\pm$ 0.01	0.77 $\pm$ 0.02	0.93 $\pm$ 0.01	0.91 $\pm$ 0.01	0.98 $\pm$ 0.01
MM-C	ACC	0.49 $\pm$ 0.02	0.5 $\pm$ 0.01	0.88 $\pm$ 0.02	0.79 $\pm$ 0.16	—	0.88 $\pm$ 0.02	0.9 $\pm$ 0.01	0.94 $\pm$ 0.01	0.95 $\pm$ 0.01	0.95 $\pm$ 0.01	0.99 $\pm$ 0.01
	Pre	0.54 $\pm$ 0.02	0.42 $\pm$ 0.05	0.87 $\pm$ 0.02	0.72 $\pm$ 0.26	—	0.88 $\pm$ 0.02	0.92 $\pm$ 0.01	0.94 $\pm$ 0.01	0.95 $\pm$ 0.01	0.96 $\pm$ 0.01	0.99 $\pm$ 0.01
	Rec	0.48 $\pm$ 0.02	0.48 $\pm$ 0.02	0.88 $\pm$ 0.02	0.74 $\pm$ 0.2	—	0.88 $\pm$ 0.02	0.91 $\pm$ 0.01	0.94 $\pm$ 0.01	0.94 $\pm$ 0.01	0.96 $\pm$ 0.01	0.99 $\pm$ 0.01
	F1	0.51 $\pm$ 0.01	0.45 $\pm$ 0.04	0.88 $\pm$ 0.01	0.69 $\pm$ 0.26	—	0.88 $\pm$ 0.02	0.9 $\pm$ 0.01	0.94 $\pm$ 0.01	0.94 $\pm$ 0.01	0.95 $\pm$ 0.01	0.99 $\pm$ 0.01
Gossipcop	ACC	0.49 $\pm$ 0.01	0.5 $\pm$ 0.01	0.78 $\pm$ 0.01	0.68 $\pm$ 0.11	0.78 $\pm$ 0.01	0.82 $\pm$ 0.01	0.97 $\pm$ 0.01	0.84 $\pm$ 0.01	0.97 $\pm$ 0.01	0.97 $\pm$ 0.01	0.98 $\pm$ 0.01
	Pre	0.33 $\pm$ 0.08	0.49 $\pm$ 0.04	0.8 $\pm$ 0.06	0.59 $\pm$ 0.21	0.78 $\pm$ 0.01	0.82 $\pm$ 0.01	0.97 $\pm$ 0.01	0.85 $\pm$ 0.01	0.97 $\pm$ 0.01	0.97 $\pm$ 0.01	0.98 $\pm$ 0.01
	Rec	0.55 $\pm$ 0.01	0.54 $\pm$ 0.01	0.77 $\pm$ 0.02	0.66 $\pm$ 0.13	0.78 $\pm$ 0.01	0.81 $\pm$ 0.01	0.98 $\pm$ 0.01	0.84 $\pm$ 0.01	0.97 $\pm$ 0.01	0.98 $\pm$ 0.01	0.98 $\pm$ 0.01
	F1	0.41 $\pm$ 0.06	0.52 $\pm$ 0.02	0.79 $\pm$ 0.02	0.59 $\pm$ 0.2	0.78 $\pm$ 0.01	0.81 $\pm$ 0.01	0.97 $\pm$ 0.01	0.84 $\pm$ 0.01	0.97 $\pm$ 0.01	0.97 $\pm$ 0.01	0.98 $\pm$ 0.01

3) *Experimental Settings and Implimentation*: For all models, the train-validation-test split of four datasets is 70%-10%-20%. The experimental results are averaged over 5-fold cross validation. We implement our method with PyTorch, and conduct experiments with Python 3.8, and 1 NVIDIA Volta GPU with 30G DRAM. In the parameter settings of ComE-DeFake, we set the hidden layer to 1 and the hidden dimension to 768. We set $\alpha = 0.001$ and $\beta = 10$ to control the balance of loss functions. The number of epochs and the number of communities are set to 400 and 2, respectively. For model optimization, we adopt the Adam to optimize the loss function by setting the learning rate to 0.01, and dropout to 0.5.

B. Overall Performance (Q1)

The results of ComE-DeFake and ten popular baseline methods are presented in TABLE II, with the optimal results highlighted in gray. ComE-DeFake achieves optimal performance on all datasets even without news text analysis, this illustrates ComE-DeFake’s superiority which learns sufficient information from news high-order relations and user credibility compared to other methods for fake news detection. Overall, ComE-DeFake confirms that in cases where the relations between news and users are fully constructed, debunking fake news without utilizing its text is feasible.

Regarding the baseline performances, the overall results of HGFND and HG-SL are not as good as those of ComE-DeFake but better than those of most other baseline methods, because these methods construct hypergraphs of social contexts as well to capture high-order relations when detecting fake news. DualEmo’s performance verifies that considering dual emotion between publishers and users, is valuable for distinguishing between real and fake news. HyperGAT also employs hypergraphs, but it builds hypergraphs of words in a piece of news, which leads to poorer performance than our method due to ignoring social contexts. HyperGAT does not work on MM-C (marked as “—” in TABLE II) because the text length of this dataset does not meet the requirements of

HyperGAT. TextGCN builds graphs at sentence level, but with short texted tweets it fails to learn enriched information about the structure among sentences, resulting in mediocre results.

The aforementioned five baseline methods incorporate structural or social context information, while the rest of the baselines only consider textual information of news. BERTweet and Llama 2 achieved performances that are only inferior to our method and other hypergraph-based baseline algorithms. This is attributed to BERTweet being pre-trained on a large corpus of tweets, and our tweet data aligns well with this model. Llama2, as the latest generative large language model, demonstrates a superior understanding of the semantics of news. Another baseline algorithm with stable performance is BERT since BERT learns effective textual features of news. TextCNN and HAN, as they do not learn structural information, are not pre-trained in the large-scale corpus, and do not consider social contexts, they have the least acceptable performance. However, the performance of these text-based baseline methods in fake news detection is worse than that of ComE-DeFake, which reflects that only learning textual information and ignoring social context leads to limited performance.

C. Ablation Study (Q2)

We perform ablation studies by discarding some important components of our best-performed approach ComE-DeFake, which include 1) *w/o HE&Clus*: We encode the user credibility information without hyperedge learning on news and community self-optimization of users. 2) *w/o HE*: We discard the hyperedge learning of news. 3) *w/o HN&Clus*: We only learn news relations on hyperedges without user credibility and community self-optimization. 4) *w/o Clus*: model without community self-optimization for users. 5) ComE-DeFake: final model with all components. Three situations of *w Text* indicate that hyperedge learning utilizes textual embeddings of news from language model [48], instead of the identity matrix. *w Attention* represents that user-to-news relation augmentation exploits attention mechanism instead of mean pooling.

TABLE III
RESULTS OF ABLATION STUDY.

Models	PolitiFact		ReCOvery		MM-C		Gossipcop	
	ACC	Rec	ACC	Rec	ACC	Rec	ACC	Rec
w/o HE&Clus	0.52 $\pm$ 0.04	0.5 $\pm$ 0.01	0.67 $\pm$ 0.02	0.5 $\pm$ 0.002	0.49 $\pm$ 0.03	0.49 $\pm$ 0.03	0.47 $\pm$ 0.1	0.46 $\pm$ 0.08
w/o HE	0.5 $\pm$ 0.09	0.47 $\pm$ 0.06	0.68 $\pm$ 0.03	0.53 $\pm$ 0.03	0.49 $\pm$ 0.04	0.5 $\pm$ 0.04	0.54 $\pm$ 0.07	0.52 $\pm$ 0.06
w/o HN&Clus	0.91 $\pm$ 0.03	0.9 $\pm$ 0.03	0.9 $\pm$ 0.02	0.86 $\pm$ 0.03	0.81 $\pm$ 0.19	0.82 $\pm$ 0.19	0.9 $\pm$ 0.06	0.89 $\pm$ 0.06
-w Text	0.91 $\pm$ 0.03	0.89 $\pm$ 0.03	0.9 $\pm$ 0.03	0.86 $\pm$ 0.04	0.89 $\pm$ 0.09	0.9 $\pm$ 0.09	0.9 $\pm$ 0.11	0.89 $\pm$ 0.12
w/o Clus	0.94 $\pm$ 0.01	0.94 $\pm$ 0.01	0.94 $\pm$ 0.08	0.9 $\pm$ 0.13	0.94 $\pm$ 0.02	0.94 $\pm$ 0.02	0.94 $\pm$ 0.02	0.94 $\pm$ 0.02
-w Text	0.93 $\pm$ 0.02	0.92 $\pm$ 0.02	0.93 $\pm$ 0.02	0.9 $\pm$ 0.03	0.93 $\pm$ 0.07	0.93 $\pm$ 0.08	0.94 $\pm$ 0.01	0.94 $\pm$ 0.02
ComE-DeFake	0.98 $\pm$ 0.03	0.98 $\pm$ 0.04	0.98 $\pm$ 0.01	0.97 $\pm$ 0.02	0.99 $\pm$ 0.01	0.99 $\pm$ 0.01	0.98 $\pm$ 0.01	0.98 $\pm$ 0.01
-w Text	0.97 $\pm$ 0.02	0.97 $\pm$ 0.02	0.97 $\pm$ 0.02	0.96 $\pm$ 0.03	0.97 $\pm$ 0.01	0.98 $\pm$ 0.01	0.96 $\pm$ 0.02	0.96 $\pm$ 0.02
-w Atten	0.98 $\pm$ 0.02	0.98 $\pm$ 0.02	0.98 $\pm$ 0.02	0.96 $\pm$ 0.03	0.99 $\pm$ 0.02	0.99 $\pm$ 0.02	0.94 $\pm$ 0.04	0.93 $\pm$ 0.04

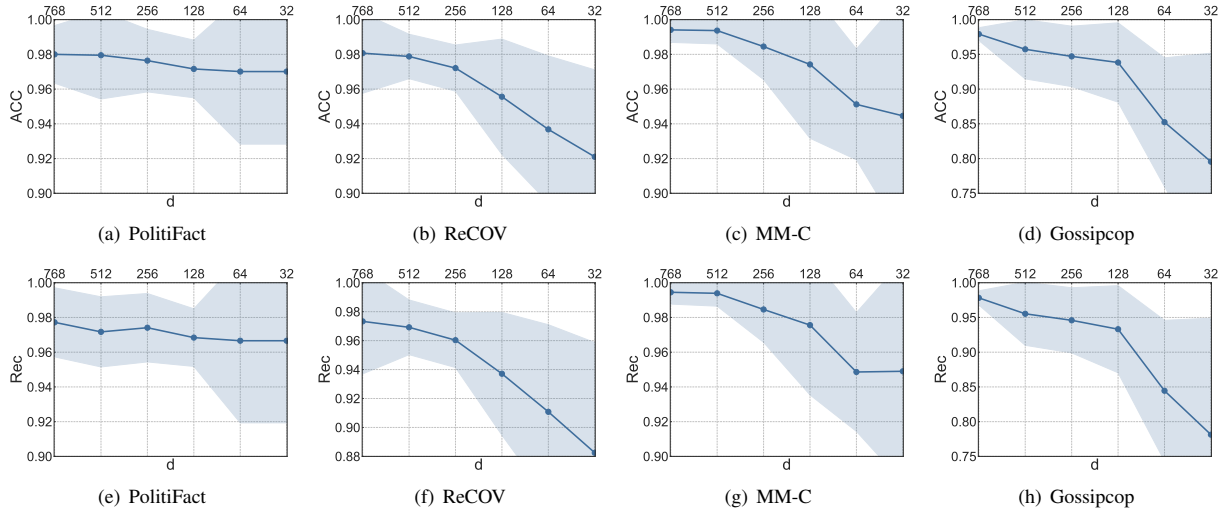


Fig. 4. Hyperparameter Analysis.

The results in TABLE III show that “w/o HE&Clus” and “w/o HE” perform the worst. Because they disregard the hyperedge learning loss function, the model loses the objective of classifying news during training and the users play a limited role. When relying solely on news, the results of “w/o HN&Clus” outperform text-based language models in TABLE II, because it can perceive the global high-order relation of news. When considering user credibility with hyperedge learning for news, the direct combination of news and user representations in “w/o Clus” shows a significant improvement compared to “w/o HN&Clus”. This implies that incorporating social contexts is crucial for superior performance. By introducing community self-optimization when capturing user credibility, our final model ComE-DeFake exhibits exceptional improvements. This indicates that ComE-DeFake, by clustering users into credible and suspicious communities, facilitates the learning of higher-order interaction between users. As a result, news representations augmented by user credibility become more discriminative. Three situations of “w Text” all yield similar or slightly worse outcomes to the models without utilizing news text due to the insufficient or fuzzy semantic

information in news, which interferes with structural relations. “w Attention” performs similarly or slightly worse than mean pooling, this may be due to the fact that the nodes in the hypergraph have learned community-level relations rather than considering individual users’ weights.

D. Sensitivity Analysis (Q3)

To investigate the sensitivity of our method on limited training labels and different hyperparameters, we vary the training label percentage and the hidden embedding size.

1) *Percentage of training data*: We adjust the percentage of training labels from 70 % to 10%. The outcomes are shown in TABLE IV, which illustrates the robust performance of ComE-DeFake on the limited number of training labels.

2) *Hyperparameter Analysis*: We evaluate the performance of ComE-DeFake with different hidden embedding sizes d of nodes and hyperedges, as shown in Fig. 4. Through the results, we set 768 as the hidden layer dimension, as our model performs best on it. When the dimension increases or decreases, our model generates stable results and outperforms the baseline methods.

TABLE IV
COME-DeFAKE'S TEST RESULTS BY VARYING THE PERCENTAGE OF AVAILABLE LABELS FOR TRAINING DATA.

Data	Metric	Percentage			
		70%	50%	30%	10%
PolitiFact	ACC	0.98 $\pm$ 0.03	0.98 $\pm$ 0.02	0.98 $\pm$ 0.02	0.97 $\pm$ 0.02
	Pre	0.99 $\pm$ 0.02	0.98 $\pm$ 0.02	0.98 $\pm$ 0.01	0.98 $\pm$ 0.02
	Rec	0.98 $\pm$ 0.04	0.98 $\pm$ 0.02	0.97 $\pm$ 0.02	0.97 $\pm$ 0.02
	F1	0.98 $\pm$ 0.03	0.98 $\pm$ 0.02	0.98 $\pm$ 0.02	0.97 $\pm$ 0.02
ReCOV	ACC	0.98 $\pm$ 0.01	0.98 $\pm$ 0.01	0.98 $\pm$ 0.02	0.97 $\pm$ 0.01
	Pre	0.98 $\pm$ 0.01	0.98 $\pm$ 0.01	0.97 $\pm$ 0.02	0.98 $\pm$ 0.01
	Rec	0.97 $\pm$ 0.02	0.96 $\pm$ 0.02	0.98 $\pm$ 0.02	0.96 $\pm$ 0.02
	F1	0.98 $\pm$ 0.01	0.97 $\pm$ 0.01	0.98 $\pm$ 0.02	0.97 $\pm$ 0.02
MM-C	ACC	0.99 $\pm$ 0.01	0.98 $\pm$ 0.03	0.98 $\pm$ 0.01	0.98 $\pm$ 0.01
	Pre	0.99 $\pm$ 0.01	0.98 $\pm$ 0.02	0.98 $\pm$ 0.01	0.98 $\pm$ 0.01
	Rec	0.99 $\pm$ 0.01	0.98 $\pm$ 0.03	0.98 $\pm$ 0.01	0.98 $\pm$ 0.01
	F1	0.99 $\pm$ 0.01	0.98 $\pm$ 0.03	0.98 $\pm$ 0.01	0.98 $\pm$ 0.01
Gossipcop	ACC	0.98 $\pm$ 0.01	0.97 $\pm$ 0.02	0.95 $\pm$ 0.04	0.95 $\pm$ 0.02
	Pre	0.98 $\pm$ 0.01	0.97 $\pm$ 0.02	0.96 $\pm$ 0.03	0.95 $\pm$ 0.02
	Rec	0.98 $\pm$ 0.01	0.97 $\pm$ 0.02	0.95 $\pm$ 0.04	0.95 $\pm$ 0.02
	F1	0.98 $\pm$ 0.01	0.97 $\pm$ 0.02	0.95 $\pm$ 0.04	0.95 $\pm$ 0.02

E. Discussion and Visualization (Q4)

To explain ComE-DeFake's superior performance without analyzing news text, we conduct visualization on four datasets. The t-SNE visualization in Fig. 5 displays the learned representations of news without using its text, demonstrating that real and fake news have different structural embeddings in low-dimensional space. The user interaction in Fig. 6 illustrates the structural relations between users involved in real and fake news. If two users participate in the same news, they are connected by an edge. We observe that users engaged in real and fake news belong to two distinct communities. These findings indicate that both real and fake news, as well as trustworthy and suspicious users, exhibit noticeable structural differences. Moreover, a positive correlation exists between news authenticity and user credibility. This phenomenon explains why we can effectively classify news even without relying on its text.

TABLE V
PERFORMANCE OF COME-DeFAKE w/o USER ATTRIBUTES.

Data	ACC	Pre	Rec	F1
PolitiFact	0.98 $\pm$ 0.02	0.99 $\pm$ 0.01	0.98 $\pm$ 0.02	0.98 $\pm$ 0.02
ReCOV	0.98 $\pm$ 0.01	0.98 $\pm$ 0.01	0.96 $\pm$ 0.02	0.97 $\pm$ 0.01
MM-C	0.99 $\pm$ 0.01	0.99 $\pm$ 0.01	0.99 $\pm$ 0.01	0.99 $\pm$ 0.01
Gossipcop	0.99 $\pm$ 0.003	0.99 $\pm$ 0.003	0.99 $\pm$ 0.003	0.99 $\pm$ 0.003

To further investigate how fake news detection benefits from user credibility, we replaced user attributes in ComE-DeFake with an identity matrix. This helps us determine which is more significant: credibility-relevant attributes (*e.g.*, verification status) or higher-order user structures. The results in TABLE V show that once our model learns high-order structural relations among users, it can discern user credibility without explicitly capturing relevant user attributes for fake news detection.

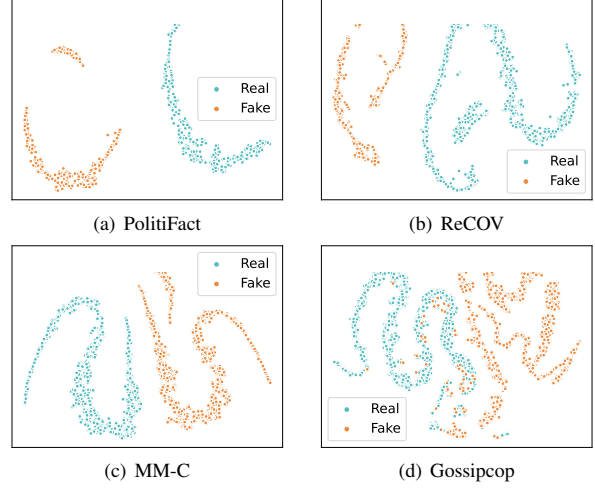


Fig. 5. Visualization of News t-SNE on Four Datasets.

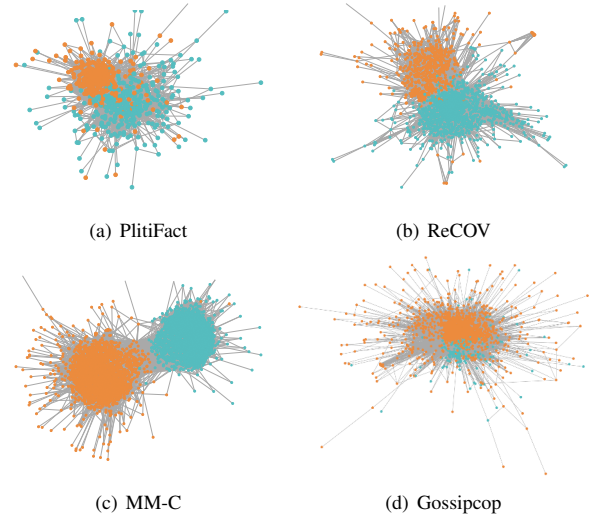


Fig. 6. User Interaction Graph (Blue nodes represent users who spread real news, while orange nodes represent users who spread fake news).

F. Early Detection (Q5)

To verify the applicability of ComE-DeFake in news spreading, especially in early detection scenarios, we evaluate its performance at intervals from 2 hours to 72 hours after news publication. The results are shown in Fig. 7, where ComE-DeFake exhibits satisfactory performance, even within the first 2 hours. This is because the formation of small-scale community structures, even with a limited number of participating users, allows our method to effectively capture high-order relations on a small scale. The experimental results validate the effectiveness of our method in detecting fake news at the early stages of news propagation, thereby mitigating the negative impact of fake news dissemination.

TABLE VI
TIME EFFICIENCY PERFORMANCE MEASURED BY CPU (C) OR GPU (G) TIME OF MODEL LEARNING IN SECONDS FOR ONE TRAINING EPOCH.

Methods	TextCNN (C)	HAN (C)	BERT (G)	TextGCN (G)	HyperGAT (G)	DualEmo (G)	HGFND (G)	BERTweet (G)	Llama (G)	HG-SL (G)	ComE-DeFake (G)
PolitiFact	10.3	40.6	1.2	0.01	7.4	0.1	12.3	2.1	4.2	1.2	0.08
ReCOV	33.4	136.1	4.6	0.03	22.8	0.4	16.1	6.8	20.9	1.4	0.27
MM-C	67	290.6	26.9	0.09	—	0.3	10.2	30.5	106.4	1	0.19
Gossipcop	164.5	697.4	98.4	0.5	103.5	2.2	33.4	143.8	522.5	2.8	2.01

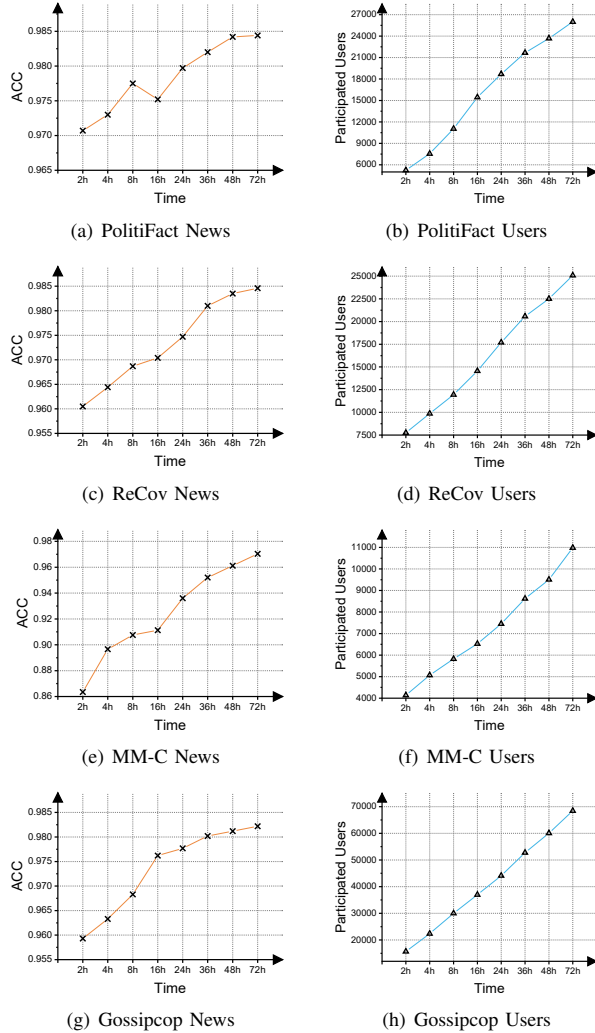


Fig. 7. Early Detection of Fake News.

G. Time Efficiency (Q6)

To assess the execution time and efficiency of ComE-DeFake, we record its training time on four datasets compared with baseline algorithms. TABLE VI presents the average training time per epoch for each method. The results demonstrate that ComE-DeFake achieves both efficient and effective training on four datasets.

VII. LOSS CONVERGENCE

To examine the convergence speed of our proposed ComE-DeFake, we illustrate the evolution of the total loss function values with epochs in Fig. 8. We can see that the convergence speeds vary among the four datasets. However, all datasets demonstrate convergence within 400 epochs.

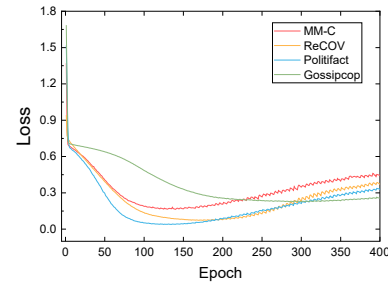


Fig. 8. Loss Convergence

VIII. CONCLUSION

To debunk online fake news, this paper argues that the existing methods, which rely heavily on semantic information, bring challenges to detecting short news or the ones requiring domain-specific knowledge. To address these challenges, our aim is to investigate the role of social relations in fake news detection and determine the feasibility of uncovering fake news without text analysis. We first construct a hypergraph of news and users, and then propose a community-driven hypergraph learning method named ComE-DeFake to achieve our goals. Extensive experiments demonstrate the superior performance of ComE-DeFake, indicating that learning distinctive embeddings that contain high-order correlations of news and users is critical in fake news detection. It addresses the limitation of the existing methods that capture only pairwise relationships between users and news. Furthermore, our findings reveal an intriguing fact, when the high-order relations between users and news are fully captured, it is feasible to classify real and fake news without considering the news text.

In our future work, we will further investigate the significance of structural relations in fake news detection, and explore the utilization of dynamic relations between news and users in fake news spreading, to suppress propagation in its early stages. Furthermore, the issue of noise caused by credible users spreading fake news due to bias or unintentional dissemination will also be explored.

REFERENCES

- [1] K. Shu, A. Sliva, S. Wang, J. Tang, and H. Liu, "Fake news detection on social media: A data mining perspective," *SIGKDD Explor.*, vol. 19, no. 1, pp. 22–36, 2017.
- [2] B. Guo, Y. Ding, L. Yao, Y. Liang, and Z. Yu, "The future of false information detection on social media: New perspectives and trends," *CSUR*, vol. 53, no. 4, pp. 1–36, 2020.
- [3] V. Pérez-Rosas, B. Kleinberg, A. Lefevre, and R. Mihalcea, "Automatic detection of fake news," in *COLING*, 2018, pp. 3391–3401.
- [4] K. Shu, L. Cui, S. Wang, D. Lee, and H. Liu, "dEFEND: Explainable fake news detection," in *KDD*, 2019, p. 395–405.
- [5] D. Rao, X. Miao, Z. Jiang, and R. Li, "STANKER: Stacking network based on level-grained attention-masked BERT for rumor detection on social media," in *EMNLP*, 2021, pp. 3347–3363.
- [6] J. Ma, W. Gao, and K.-F. Wong, "Rumor detection on Twitter with tree-structured recursive neural networks," in *ACL*, 2018, pp. 1980–1989.
- [7] T. Bian, X. Xiao, T. Xu, P. Zhao, W. Huang, Y. Rong, and J. Huang, "Rumor detection on social media with bi-directional graph convolutional networks," in *AAAI*, 2020, pp. 549–556.
- [8] L. Cui, H. Seo, M. Tabar, F. Ma, S. Wang, and D. Lee, "Deterrent: Knowledge guided graph attention network for detecting healthcare misinformation," in *KDD*, 2020, p. 492–502.
- [9] N. Rosenfeld, A. Szanto, and D. C. Parkes, "A kernel of truth: Determining rumor veracity on twitter by diffusion pattern alone," in *WWW*, 2020, p. 1018–1028.
- [10] X. Su, J. Yang, J. Wu, and Y. Zhang, "Mining user-aware multi-relations for fake news detection in large scale online social networks," in *WSDM*, 2023, p. 51–59.
- [11] K. Shu, S. Wang, and H. Liu, "Beyond news contents: The role of social context for fake news detection," in *WSDM*, 2019, p. 312–320.
- [12] A. Bretto, "Hypergraph theory," *An introduction. Mathematical Engineering*. Cham: Springer, 2013.
- [13] J. Zhou, G. Cui, S. Hu, Z. Zhang, C. Yang, Z. Liu, L. Wang, C. Li, and M. Sun, "Graph neural networks: A review of methods and applications," *AI open*, vol. 1, pp. 57–81, 2020.
- [14] Y. Feng, H. You, Z. Zhang, R. Ji, and Y. Gao, "Hypergraph neural networks," in *AAAI*, 2019, pp. 3558–3565.
- [15] N. Yadati, M. Nimishakavi, P. Yadav, V. Nitin, A. Louis, and P. Talukdar, "Hypergcnn: A new method for training graph convolutional networks on hypergraphs," in *NIPS*, 2019.
- [16] R. Zhang, Y. Zou, and J. Ma, "Hyper-sagnn: a self-attention based graph neural network for hypergraphs," *arXiv preprint arXiv:1911.02613*, 2019.
- [17] T. Wei, Y. You, T. Chen, Y. Shen, J. He, and Z. Wang, "Augmentations in hypergraph contrastive learning: Fabricated and generative," in *NIPS*, 2022, pp. 1909–1922.
- [18] H. Wu, Y. Yan, and M. K.-P. Ng, "Hypergraph collaborative network on vertices and hyperedges," *TPAMI*, vol. 45, no. 3, pp. 3245–3258, 2022.
- [19] K. Ding, J. Wang, J. Li, D. Li, and H. Liu, "Be more with less: Hypergraph attention networks for inductive text classification," in *EMNLP*, 2020, pp. 4927–4936.
- [20] U. Jeong, K. Ding, L. Cheng, R. Guo, K. Shu, and H. Liu, "Nothing stands alone: Relational fake news detection with hypergraph neural networks," in *Big Data*, 2022, pp. 596–605.
- [21] L. Sun, Y. Rao, Y. Lan, B. Xia, and Y. Li, "Hg-sl: Jointly learning of global and local user spreading behavior for fake news early detection," in *AAAI*, 2023, pp. 5248–5256.
- [22] R. K. Kaliyar, A. Goswami, and P. Narang, "Fakebert: Fake news detection in social media with a bert-based deep learning approach," *Multimed. Tools. Appl.*, vol. 80, no. 8, pp. 11 765–11 788, 2021.
- [23] X. Zhang, J. Cao, X. Li, Q. Sheng, L. Zhong, and K. Shu, "Mining dual emotion for fake news detection," in *WWW*, 2021, p. 3465–3476.
- [24] R. Yang, J. Ma, H. Lin, and W. Gao, "A weakly supervised propagation model for rumor verification and stance detection with multiple instance learning," in *SIGIR*, 2022, p. 1761–1772.
- [25] Q. Sheng, J. Cao, X. Zhang, R. Li, D. Wang, and Y. Zhu, "Zoom out and observe: News environment perception for fake news detection," in *ACL*, 2022, pp. 4543–4556.
- [26] S. Gong, R. O. Sinnott, J. Qi, and C. Paris, "Fake news detection through graph-based neural networks: A survey," *arXiv preprint arXiv:2307.12639*, 2023.
- [27] J. Wu, Q. Liu, W. Xu, and S. Wu, "Bias mitigation for evidence-aware fake news detection by causal intervention," in *SIGIR*, 2022, p. 2308–2313.
- [28] L. Yao, C. Mao, and Y. Luo, "Graph convolutional networks for text classification," in *AAAI*, 2019, pp. 7370–7377.
- [29] Y. Jin, X. Wang, R. Yang, Y. Sun, W. Wang, H. Liao, and X. Xie, "Towards fine-grained reasoning for fake news detection," in *AAAI*, 2022, pp. 5746–5754.
- [30] V.-H. Nguyen, K. Sugiyama, P. Nakov, and M.-Y. Kan, "FANG: leveraging social context for fake news detection using graph representation," in *CIKM*, 2022, p. 124–132.
- [31] L. Hu, T. Yang, L. Zhang, W. Zhong, D. Tang, C. Shi, N. Duan, and M. Zhou, "Compare to the knowledge: Graph neural fake news detection with external knowledge," in *ACL/IJCNLP*, 2021, pp. 754–763.
- [32] M. Mayank, S. Sharma, and R. Sharma, "DEAP-FAKED: Knowledge graph based approach for fake news detection," in *ASONAM*, 2022, pp. 47–51.
- [33] Y. Ren and J. Zhang, "Fake news detection on news-oriented heterogeneous information networks through hierarchical graph attention," in *IJCNN*, 2021, pp. 1–8.
- [34] R. Yang, X. Wang, Y. Jin, C. Li, J. Lian, and X. Xie, "Reinforcement subgraph reasoning for fake news detection," in *KDD*, 2022, p. 2253–2262.
- [35] N. Mehta, M. L. Pacheco, and D. Goldwasser, "Tackling fake news detection by continually improving social context representations using graph neural networks," in *ACL*, 2022, pp. 1363–1380.
- [36] Y. Dou, K. Shu, C. Xia, P. S. Yu, and L. Sun, "User preference-aware fake news detection," in *SIGIR*, 2021, pp. 2051–2055.
- [37] E. Min, Y. Rong, Y. Bian, T. Xu, P. Zhao, J. Huang, and S. Ananidou, "Divide-and-conquer: Post-user interaction network for fake news detection on social media," in *WWW*, 2022, p. 1148–1158.
- [38] X. Su, S. Xue, F. Liu, J. Wu, J. Yang, C. Zhou, W. Hu, C. Paris, S. Nepal, D. Jin, Q. Z. Sheng, and P. S. Yu, "A comprehensive survey on community detection with deep learning," *TNNLS*, pp. 1–21, 2022.
- [39] K. Shu, D. Mahudeswaran, S. Wang, D. Lee, and H. Liu, "Fakenewsnet: A data repository with news content, social context, and spatiotemporal information for studying fake news on social media," *Big data*, vol. 8, no. 3, pp. 171–188, 2020.
- [40] X. Zhou, A. Mulay, E. Ferrara, and R. Zafarani, "ReCOVery: A multimodal repository for covid-19 news credibility research," in *CIKM*, 2020, pp. 3205–3212.
- [41] Y. Li, B. Jiang, K. Shu, and H. Liu, "Mm-covid: A multilingual and multimodal data repository for combating covid-19 disinformation," 2020.
- [42] Y. Kim, "Convolutional neural networks for sentence classification," in *EMNLP*, 2014, pp. 1746–1751.
- [43] Z. Yang, D. Yang, C. Dyer, X. He, A. Smola, and E. Hovy, "Hierarchical attention networks for document classification," in *NAACL-HLT*, 2016, pp. 1480–1489.
- [44] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *NAACL-HLT*, 2019, pp. 4171–4186.
- [45] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *ICLR*, 2017.
- [46] D. Q. Nguyen, T. Vu, and A. Tuan Nguyen, "BERTweet: A pre-trained language model for English tweets," in *EMNLP*, 2020, pp. 9–14.
- [47] H. Touvron, L. Martin, K. Stone, P. Albert, A. Almahairi, Y. Babaei, N. Bashlykov, S. Batra, P. Bhargava, S. Bhosale et al., "Llama 2: Open foundation and fine-tuned chat models, 2023," URL <https://arxiv.org/abs/2307.09288>, 2023.
- [48] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, "RoBERTa: A robustly optimized BERT pretraining approach," *arXiv preprint arXiv:1907.11692*, 2019.